



UNSW SYDNEY

UNSW Business School
School of Risk and Actuarial Studies

ACTL3142/5110: Statistical Machine Learning for Risk and Actuarial Applications

Assessment Task	Assignment Part 1
Due date	Friday 3rd July 2026, 11:55am
Weight (%)	12%
Task description	Individual Assignment
Submission Method	Moodle Turnitin
Feedback mechanisms	Breakdown of marks per question, individualised comments & global feedback with “common mistakes”.
Links to Unit Learning Outcomes	CLO1, CLO2, CLO3, CLO5

INTEGRITY MATTERS

As a valued member of the UNSW community, you have the responsibility to uphold the values found in its Code of Conduct: integrity, honesty and trust. After all, UNSW's reputation is *your* reputation. This is important because our education system is built on trust and merit. Students can *trust* that their lecturers will grade them fairly and this gives *merit* to their grades. When students cheat, it erodes the trust people have in this system, hence weakens an institution that is a pillar of our society. Upon completing your degree, society expects you to apply the highest standards of ethics in your work. It is a good idea to start practicing today.

If you are ever unsure whether your actions fall within the guidelines of UNSW's Academic Integrity, please don't hesitate to reach out to your lecturer in charge.

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1. Context

In this assignment, you will continue analysing the `airbnb-Sydney.csv` data set, which should be familiar to you because of the formative tasks “StoryWall 1 & 2”. The data set is found under Ed Resources, and contains information about 8,000 Airbnb listings in Sydney, Australia. The file `AirbnbDataDictionary.xlsx` contains a description of most variables in the data set. The descriptions are brief – if you are unsure about a variable’s meaning, feel free to ask on Ed Discussion.

For this Assignment, we are interested in understanding the `price_per_room` of a given Airbnb listing as a function of relevant predictor variables under both a standard linear regression model and a GLM. In terms of structure, `price_per_room` will be considered the **response** variable, and we henceforth denote it “ Y ”. As mentioned in the StoryWalls, `price_per_room` represents the daily price of an Airbnb listing divided by the total number of bedrooms available to the customer booking the Airbnb. If the number of bedrooms is “0”, then `price_per_room` is just equal to “daily price”. The other variables in this data set are viewed as potential predictors – note that not every predictor may be useful.

For the following tasks, assume that you are an analyst working for a private investor who is looking to buy properties and potentially list them on Airbnb to generate rental income.

You should answer all the questions in the next section in a well organised report, in PDF format (see also Section 3: Directives and Format Requirements).

2. Your Tasks

1. (2.25pt) Conduct an exploratory data analysis to find a few predictors that appear to be meaningfully related to the variable “ $Y := \text{price_per_room}$ ”. Present data summaries and visualisations which clearly illustrate the nature of the relationships between these predictors and the response, and interpret these relationships (i.e., explain intuitively why these relationships exist). Your analysis should include at least three predictors, of which at least one is numerical and at least one is qualitative (“categorical”).

Note: You should view this step as preliminary to the creation of the regression models, as it serves to establish which predictors are likely to be the most important in your models.

2. (2.25pt) Create the best linear regression model that you can to explain Y via the predictors in the data set. You are free to use any of the variables in the data set

as predictors, transforming the variables and/or creating new ones from the existing ones and/or including interactions between variables. **However, you are not allowed to import new variables from external sources.** Present a summary of your model (containing all the information needed to interpret your model). Carefully explain and justify what makes it the best model you could create.

Note: You are free to choose which criteria to use when deciding which model is “the best”, but you need to clearly explain your rationale.

3. (2.25pt) Create the best Generalised Linear Model that you can to explain Y via the predictors (the same restrictions as in Question 2 apply). Present a summary of your model (containing all the information needed to interpret your model). Carefully explain and justify what makes it the best model you could create.

Note: Again, you are in charge of choosing what makes a model “the best” but you need to clearly explain your rationale. Also, remember that with GLMs, you have more “choices” than in linear regression. For example, you will need to choose the stochastic component and the link function. Those choices should be justified.

4. (2.25pt) Because any linear regression model (under the strong assumptions) is a “special case” within the GLM framework, your model from Question 3 should be *no worse* than your model from Question 2. Use appropriate diagnostic tools and arguments to explain:
 - **whether** and **to what extent** your GLM model improves upon your linear regression model.
 - whether there are any **material issues/flaws** remaining with your GLM model.
5. (1.75pt) Based on all your findings from the previous questions, describe the relationships you have uncovered between `price_per_room` and your selected predictors. In particular, what useful insights do your results provide to an investor looking to buy properties and list them on Airbnb for the purpose of generating rental income?

3. Directives and Format Requirements

All students must submit an assignment of their **individual own work**. Students must comply with the following directives and format requirements. Penalties will be applied to any breach of such directives (and we will not grade any work that exceeds the page limit).

- You must submit your assignment via Turnitin (on **Moodle**), and you have **2 distinct files** to submit:
 - a **.pdf** file: this is your report.
 - a file with all your computer code.
- Only **typed assignments (not handwritten)** will be accepted.

- The .pdf file should include a title page with: student name, student ID.
- The **main body** of your .pdf report (excluding title page, appendices, bibliography) must not exceed **4 pages**. It must contain everything that is asked in the questions (e.g., results, explanations, plots, diagnostics, etc.), but should not include your computer code. For Q2 and Q3, you are permitted to place your “model summaries” in an Appendix. Where relevant, please show your reasoning and thought process as clearly as possible in your answers.
- You must **number** your answers, in the same way the questions are numbered.
- Your report must include an appendix titled “Generative AI usage” explaining what you used AI for in this assignment. See Section 4.2 for more details.
- All computer code necessary to produce your results **must ALSO be placed in an Appendix** to your report. There is no page limit for this Appendix.
- This code within the appendix must be made of **text** (not images), that we can copy-paste.
- Your code in the **code file** must run “as is” (and produce exactly the results in your assignment). If we cannot run your code, you will lose ALL marks attached to results this code is meant to produce.
- The page format to use is A4 (this is the standard Australian format).
- The minimum font size must be an equivalent of ‘Times New Roman’ size 11.
- The minimum line spacing to use is 1.15.
- The margins must not be narrower than the ‘narrow’ option in Word (which is 0.5 inches every side).
- Your code should also include the relevant question numbers (as comments, e.g., #Q1, #Q2, etc.).

4. Marking Guidelines

4.1 Marking Criteria

Individual questions are allocated a fixed number of marks, and will be evaluated using Criteria C1 and C2 as defined below (the descriptions provided correspond to a “HD mark”). The overall “Presentation and Professionalism” of your report also counts for 1.25/12 of the mark, and will be assessed under criteria C3 below. Lastly, we will deduct marks for any breach of the instructions & requirements detailed in Section 3.

- **C1: Methods and Results.** Chooses and correctly applies appropriate methods to derive relevant results. Provides comprehensive and compelling reasoning to justify all components of these methods. Where relevant, uses correct diagnostic tools to assess the appropriateness and/or performance of the methods used. Demonstrates that many different models/methods have been considered, and that the best ones have been selected based on sensible criteria. Calculations, computer codes and other workings produce results which are correct and answer the questions asked. Does not produce any incorrect, irrelevant or superfluous results.
- **C2: Analysis and Communication.** Performs insightful and accurate analysis of the results obtained. Interprets findings correctly and precisely. Shows a great depth of understanding and critical thinking when interpreting/analysing findings. Accurately identifies and explains the assumptions and limitations of the methods used. Explains the relevant nuances and caveats of the results obtained, where necessary. Expresses all ideas clearly and with focus, with no irrelevant or repetitive material. Develops ideas in a logical and coherent manner. Presents opinions and arguments logically and persuasively. Uses fluent, precise and concise language which is appropriate for the intended audience, context and purpose.
- **C3: Presentation and Professionalism.** Presents all findings very professionally. Shows evidence of thorough editing (e.g., no typos, grammar, spelling or punctuation errors). Chooses style(s) of presentation appropriate for the specific business/academic context. Formats and presents results clearly and neatly. Displays figures and tables in a readable and aesthetic format. Chooses appropriate visualisation and/or data summary methods that convey all (and only) the appropriate information.

4.2 Plagiarism Awareness and AI Use

This is an individual assignment. The material each student submits must be their own individual work. Students should make sure they understand what plagiarism is. Note that plagiarism rules also apply to your computer code, which must be created from your own device, and developed by you. The best strategy to avoid any problem is to never share pieces of your work with other students.

You are allowed to use Generative Artificial Intelligence (AI) to help you with editing, planning, idea generation, or coding. However, you must include an Appendix in your report titled "Generative AI usage" explaining what you used AI for and, if applicable, outlining what prompts you used. If you did not use Generative AI, you must still write in this Appendix that generative AI was not used. Copying and pasting computer code, sentences, or paragraphs from AI tools without proper citation is considered plagiarism (in the same way it would be considered as plagiarism if you copy-pasted sentences from your friend's assignment, or any other source). When referencing AI, you must follow UNSW's official guideline as provided here: <https://www.student.unsw.edu.au/ai-referencing>.

4.3 Late Submission

Any late submission will incur a penalty of 5% per day or part thereof (including weekends) from the due date and time. An assessment will not be accepted after 5 days (120 hours) of the original deadline unless special consideration has been approved. In the case of an

approved Equitable Learning Plan (ELP) provision, special consideration or short extension, the late penalty applies from the date of approved time extension. After five days from the extended deadline, the assessment cannot be submitted.

4.4 Short Extension

“Short Extension” is a new process that allows you to apply for an extended deadline on your assessment without the need to provide supporting documentation, offering immediate approval during brief, life-disrupting events. Requests are automatically approved once submitted. For this assignment, a Short Extension of **2 days** is available. Details on how to submit a Short Extension request (or a “Special Consideration” request) are found here: <https://www.student.unsw.edu.au/special-consideration>. Note you must do this **before** the official due date of the assignment.

5. Students’ Questions

5.1 Answering Students’ Questions on Ed

Questions about the assignment must be posted on Ed Discussion, under the category “Assessments”. We do not plan to give out many additional instructions or hints, but **if** we do, it is only fair that everyone will have access to the response. Please note that there can be a delay of a few days between a question and its answer. In particular, we do not plan to answer questions posted less than 48h before the official deadline. Hence, please get your questions in by Wednesday 11:55am of Week 5.

Lastly, before posting a question, please have a look to see if your question has already been answered in an existing thread.